COMP 4531 Final Project Draft

Automating Rotary-Wing QC with CNN-based Anomaly Detection on X-ray & Ultrasound

1) Problem framing & business case

Goal: Assist NDI/NDT inspectors by flagging likely defects (voids, delaminations, porosity, cracks, disbonds, FOD) in x-ray and ultrasound images of rotary-wing structures (rotor blades, hubs, gearbox housings, spars). Why now: Throughput and first-pass yield targets demand faster, more consistent screening; missed defects are high-severity (safety, rework, MRB). Deliverable: A validated, auditable model+workflow that triages images, localizes anomalies with heatmaps/masks, produces inspector-readable evidence, and integrates into QC stations.

2) Data scope, access & governance

Modalities: X-ray (DICOM/TIFF) and ultrasound (B-scan/C-scan) images. Labeling tiers: weak (part-level), strong (mask-level), unsupervised (normal images). Splits: Stratified by part number, line, and machine; holdout blind test. Data hygiene: Normalize bit depth, clamp outliers, remove overlays, preserve raw for audit.

3) Technical approach

Three-track comparison: A. Supervised classification (ResNet/MobileNet) B. Supervised segmentation (U-Net with Dice+BCE) C. Unsupervised anomaly detection (Convolutional Autoencoder) All trained with TensorFlow/Keras per course methods (SGD/Adam, LR schedules, dropout/L2, early stopping).

4) Model & training recipes

Preprocessing: Per-image normalization, cropping, conservative augmentations. Classifier: ResNet-18 backbone with Grad-CAM explainability. Segmenter: U-Net encoder–decoder with skip connections and transposed convolutions. Autoencoder: Symmetric CNN trained with MSE+SSIM loss for structural anomalies. Optimization: Adam or SGD with cosine decay LR; regularization and class reweighting.

5) Evaluation plan

Offline metrics: PR-AUC, recall@precision, Dice/IoU, pixel-AUPRC, FN rate per part. Robustness: Domain-shift holdout, SNR stress tests, ablations. Human-in-loop pilot: Compare inspector review time, false negatives, overlay trust. Acceptance: Recall >= 0.98 for critical defects, FP rate within QC bounds.

6) Risks & mitigations

Class imbalance -> Focal loss, oversampling. Spurious cues -> Crop overlays, augment variation. Domain shifts -> Normalization, fine-tuning per station. Explainability -> Grad-CAM and segmentation overlays. Ethics & safety -> FN control, inspector remains final authority.